

Profile Summary

Junior Data Scientist & AI/ML Engineer with a strong foundation in Electronics and Embedded Systems, skilled in Python, SQL, PyTorch, and Machine Learning. Experienced in building scalable AI/ML pipelines, computer vision models, and predictive analytics solutions. Proficient in YOLO, OpenCV, SAM, data preprocessing, EDA/ETL, and model deployment on platforms like Raspberry Pi. Blend of hardware knowledge and data-driven expertise, making me well-suited for product-based innovation in AI, IoT, and automation.

Work Experience

- Python Programming Intern – Antsy

Nov 2024 – Dec 2024

 - Automated workflows and data cleaning tasks using Python, reducing manual effort and errors.
 - Performed data analysis, visualization, and reporting using Pandas, Matplotlib, and Seaborn.
 - Developed a mini ML project predicting house prices using Scikit-learn.
 - Built GUI applications with Tkinter and integrated APIs & web scraping for real-time data.
 - Collaborated using Git/GitHub to maintain code quality and version control.
- Data Science & Analytics Internships – Deloitte, BCGx, Tata (Forage)

May 2025

 - Conducted EDA, feature engineering, and model evaluation on structured datasets.
 - Developed interactive dashboards and optimized SQL queries for efficient data retrieval.
 - Applied data-driven problem-solving and analytical thinking in professional settings.

Projects & Research

- Brain Tumor Analysis & Diagnosis System | Python, YOLOv12, SAM, ResNet18 [Link](#)

Apr 2025

 - Built an end-to-end AI pipeline for MRI tumor detection: classification, detection, and segmentation.
 - Applied data augmentation and preprocessing for improved model generalization.
 - Evaluated performance using Accuracy, IoU, and Dice Coefficient.
 - Optimized deployment on Raspberry Pi for low-resource edge inference.
- Automatic Number Plate Recognition (ANPR) | Python, OpenCV, YOLOv8 [Link](#)

Jun 2024

 - Developed a real-time ANPR system achieving 93.75% accuracy.
 - Performed preprocessing to handle lighting variations and occlusion.
 - Deployed edge-AI solution using Raspberry Pi.
- English-to-SQL Generator | Python, Google Gemini API, MySQL [Link](#)

Jul 2025

 - Built AI-powered SQL query generator for dynamic database schemas.
 - Automated query generation and implemented secure database connectivity.
 - Integrated API-driven data retrieval for real-time analytics.

Technical Skills

- Programming & Tools:** Python, SQL, REST API, HTML, Git/GitHub, Jupyter Notebook, Google Colab
- Libraries & Frameworks:** PyTorch, Scikit-learn, OpenCV, YOLO, SAM, NumPy, Pandas, Matplotlib, Seaborn, Selenium
- Concepts & Methods:** Machine Learning, Deep Learning, Computer Vision, Object Detection, Image Segmentation, EDA, ETL, Feature Engineering, Predictive Modeling, Model Deployment, RAG
- Electronics:** Circuit Design, Embedded Systems, Raspberry Pi, Sensors & IoT,
- Platforms:** Raspberry Pi, Linux, Windows, Tableau
- Engineered a multi-cloud data lake architecture** using the Bronze–Silver–Gold framework for seamless data ingestion, transformation, and storage across AWS, Azure, and GCP.
- Built scalable data pipelines** for efficient processing and integration, leveraging Apache Spark, AWS Glue, and Azure Data Factory.

Certifications

- GenAI Powered Data Analytics – **Tata Group** (Forage)
 - Data Analytics Virtual Experience – **Deloitte** (Forage)
 - Data Science Intern Virtual Experience – **BCGx** (Forage)
- May 2025

Jun 2025

Jun 2025

Additional Information

- Typing: 50 WPM (Certified)**
- Soft Skills:** Analytical Thinking, Problem-solving, Team Collaboration, Technical Documentation, Effective Communication, Research & Development, Story Writing.
- Interests:** Data Science, AI & ML, Biomedical AI, Autonomous Systems, Analytics-driven problem solving

Education

B.E. – Electronics & Communication Engineering (CSE Specialization)
University College of Engineering, Kancheepuram – **Anna University**
CGPA: 7.54 / 10

2021 – 2025